Polifonia: a digital harmoniser for musical heritage knowledge, H2020

D4.4: Software for knowledge extraction from text – entities – 2nd version (V1.0)

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Project Information

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1	UNIBO	ALMA MATER STUDIORUM - UNIVERSITÀ DI BOLOGNA	Italy
2	OU	THE OPEN UNIVERSITY	United Kingdom
3	KCL	KING'S COLLEGE LONDON	United Kingdom
4	NUI GALWAY	NATIONAL UNIVERSITY OF IRELAND GALWAY	Ireland
5	MiC	MINISTERIO DELLA CULTURA	Italy
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	SORBONNE	SORBONNE UNIVERSITE (LinkedTP)	France
7	CNAM	CONSERVATOIRE NATIONAL DES ARTS ET METIERS	France
8	NISV	STICHTING NEDERLANDS INSTITUUT VOOR BEELD EN GELUID	Netherlands
9	KNAW	KONINKLIJKE NEDERLANDSE AKADEMIE VAN WETENSCHAPPEN	Netherlands
10	DP	DIGITAL PATHS	Italy

Project Summary

European musical heritage is a dynamic historical flow of experiences, leaving heterogeneous traces that are difficult to capture, connect, access, interpret, and valorise. Computing technologies have the potential to shed a light on this wealth of resources by extracting, materialising and linking new knowledge from heterogeneous sources, hence revealing facts and experiences from hidden voices of the past. Polifonia makes this happen by building novel ways of inspecting, representing, and interacting with digital content. Memory institutions, scholars, and citizens will be able to navigate, explore, and discover multiple perspectives and stories about European Musical Heritage.

Polifonia focuses on European Musical Heritage, intended as musical contents and artefacts - or music objects - (tunes, scores, melodies, notations, etc.) along with relevant knowledge about them such as their links to tangible objects (theatres, conservatoires, churches, etc.), their cultural and historical contexts, opinions and stories told by people having diverse social and artistic roles (scholars, writers, students, intellectuals, musicians, politicians, journalists, etc), and facts expressed in different styles and disciplines (memoirs, reportage, news, biographies, reviews), different languages (English, Italian, French, Spanish, and German), and across centuries.

The overall goal of the project is to realise an ecosystem of computational methods and tools supporting discovery, extraction, encoding, interlinking, classification, exploration of, and access to, musical heritage knowledge on the Web. An equally important objective is to demonstrate that these tools improve the state of the art of Social Science and Humanities (SSH) methodologies. Hence their development is guided by, and continuously intertwined with, experiments and validations performed in real-world settings, identified by musical heritage stakeholders (both belonging to the Consortium and external supporters) such as cultural institutes and collection owners, historians of music, anthropologists and ethnomusicologists, linguists, etc.

Executive Summary

Deliverable D4.4 is titled: Software for knowledge extraction from text – entities – 2nd version. The deliverable summarises information about the software components under development in the Polifonia Work Package 4 dedicated to the identification and extraction of knowledge (entities) from textual resources. It is due month 30 and linked to Task 2: Automatic extraction of time, space, events, people and musical artifacts from text. The task leader is OU (The Open University) and the participants are UNIBO and OU. This deliverable and task are strongly linked with research and development work undertaken in relation to pilots CHILD, MEETUPS, and MUSICBO, although the methods developed can be applied to other use cases as well. The resources described in this deliverable are released in the Polifonia Ecosystem.

Document History

Version	Release date	Summary of changes	Author(s) - Institution
v0.1	07/07/2023	Deliverable layout	Enrico Daga (OU)
v0.2	20/07/2022	Deliverable subsections update	Alba Morales Tirado (OU)
v0.3	21/08/2023	Section 1 and 2 completed	Alba Morales Tirado (OU)
v0.4	29/08/2023	Section 3 completed	Arianna Graciotti (UNIBO)
v0.5	31/08/2023	Section 4 completed	Jason Carvalho (OU), Alba Morales Tirado (OU)
v0.6	07/09/2023	Deliverable draft ready for reviewers	Alba Morales Tirado (OU), Jason Carvalho (OU), Enrico Daga (OU)
v0.7	25/09/2023	Revised version incorporating feedback from reviewers.	Alba Morales Tirado (OU), Enrico Daga (OU)
v1.0	02/10/2023	Submission to EU	Coordinator

Table of contents

1 Introduction

This deliverable collects the software components developed in the Polifonia Work Package 4 and describes the software artifacts built for the identification and extraction of knowledge (entities) from textual resources. Particularly, it presents the tools developed for the automatic extraction of people, places, and time expression entities, as well as the automatic classification of text according to events of interest for social network analysis in the music domain.

The software presented here is the output of Task 2: Automatic extraction of time, space, events, people and musical artifacts from text. In the previous Deliverable 4.3 [1], we presented the first outcomes of the knowledge extraction process resulting in a set of tools that support entity recognition, linking and theme classification of textual data. In this deliverable, we focused on describing new components in charge of coreference and historical meetup identification. This deliverable and task are strongly linked with research and development work undertaken concerning pilots CHILD, MUSICBO, and MEETUPS, although the methods developed can be applied to other use cases as well.

The software components presented in this deliverable are:

- **Software for the extraction of historical meetups.** Refers to the set of components implementing a knowledge extraction pipeline. Although they together implement a composite process, from the collection of textual biographies to a structured knowledge graph, the components can also be used separately, or customised for similar tasks. The components described here are:
  - **Intra-document coreference resolution.** This component provides means to find mentions that refer to the same named entities within the text. Importantly in this deliverable, we focus our attention on people and places. These mentions refer to noun phrases and pronouns that indicate a mention of a particular entity and are difficult to identify using entity recognition and linking techniques.
  - **Identification of Historical Meetups.** This component processes the results of the entity recognition step and the coreference resolution and runs an automatic evaluation of the text that contains all the constituent parts of a meetup and therefore, can be identified as a historical meetup.
  - **Software for entity recognition**. These sub-components were released in the previous Deliverable and in this report, we describe the updates in software to improve the quality of the results.
- **Polifonia lexicon annotator**. The component provides means to annotate text with the Polifonia lexicon, developed in Task 4.1.
- **Search expansion using Large Language Models (LLMs)**. This component provides a means to extract documents that refer to a given implicit theme taking as reference the documentary evidence benchmark. It uses Large Language Models (LLMs) tools to expand the search and extraction of such documents. It relies on the Listening Experience Database (LED) as the gold standard.

The resources described in this deliverable are released in the Polifonia Ecosystem version 2.0 (released in August 2023).

2 Extraction of Historical Meetups

This section describes the components developed to extract information about Historical Meetups. Section 2.1 is dedicated to giving a brief summary of the MEETUPS Pilot objectives and motivation to extract entities from text and build a dataset of historical meetups. In Section 2.2, we present a set of software tools for entity recognition and linking. Importantly, this deliverable introduces two new software components dedicated to entity coreference resolution and identification of historical meetups. Furthermore, we describe components that were updated since their release in the previous deliverable [1]. The components' releases are published on Zenodo [2] and available in the MEETUPS repository[1].

2.1 Background

This work is part of the MEETUPS pilot [2], which focuses on supporting music historians and teachers by providing a Web tool that enables the exploration and visualisation of encounters between people in the musical world in Europe from c.1800 to c.1945, relying on information extracted from public domain books such as biographies, memoirs and travel writing, and open-access databases.

Such encounters were particularly significant before the period when musical ideas and influences were widely disseminated through broadcasting and recording technologies, and the collection of data about them will make it possible to trace key points of cultural and musical exchange and dissemination and meetings that were catalysts for musical change. It may also reveal unexpected connections and relationships that cast new light on aspects of European music history. These encounters will be explored in a timeline and map interface and may reveal unexpected connections and relationships that cast new light on aspects of European music history. The tool will provide persistent, citable identifiers in order to support referencing in scholarship outputs.

As an overview, the MEETUPS Pilot is focused on:

- collecting the knowledge requirements in terms of data expected and interface to be displayed for exploration in the Web tool,
- selecting and collecting data sources according to availability and objective,
- designing the knowledge extraction pipeline in charge of gathering all the data and,
- finally building the knowledge components that are the base of the Web tool, specifically the MEETUPS Knowledge Graph [3] and MEETUPS Ontology [4].

In this deliverable, we focus on the software developed to implement the knowledge extraction pipeline. In what follows, we describe the process to define the components of a Historical Meetup (Section 2.1.1), the design of the knowledge extraction pipeline (Section 2.1.2), and the software components devised to extract the entities part of the historical meetup (Section 2.2).

2.1.1 Definition of Historical Meetups

The knowledge requirements of scholars looking to uncover such encounters are identified and stated in the form of Persona stories and formalised as Competency Questions (CQs)[2] [5]. These requirements were compiled in the Polifonia stories named Ortenz, David, and Sophia.

[1] MEETUPS repository: https://github.com/polifonia-project/meetups_pilot
[2] Persona stories and CQs: https://polifonia-project.github.io/ecosystem/pages/pilots/meetups.html

The stories describe a series of scenarios regarding users' scholarly activities, their data needs and interests in terms of music and cultural heritage, for instance, their requirements to analyse interactions between music personalities and the discovery of patterns for social exchange.

From this analysis, we identify the elements of a *historical meetup*. Four components can be listed (see Figure 2.1):

- Participant(s): people involved (who?)
- Location: the place where it took place (where?)
- Type of event: the event that took place, the reason (what?)
- Time expression: the date or moment in time when the meetup happened (when?)

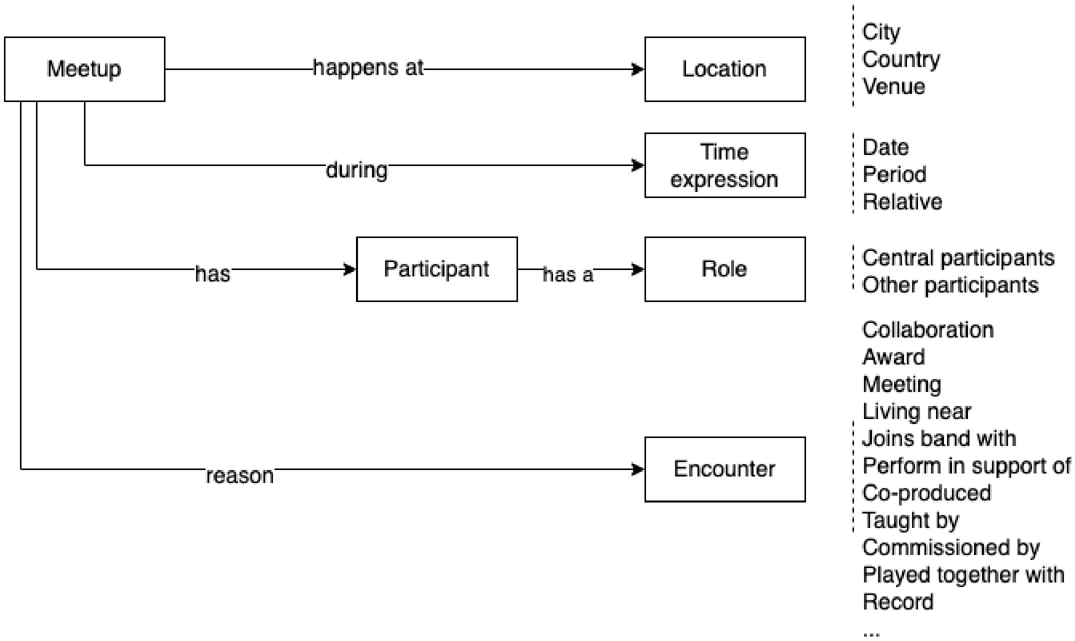

Figure 2.1: Historical Meetup definition

These elements should be identified and extracted from public domain sources such as books, memoirs or biographies. Furthermore, and when possible they should be linked to databases such as DBpedia or Wikidata. We approach this problem from the perspective of knowledge extraction tasks such as Named Entity Recognition and Linking (NER) in order to obtain the elements shown in Figure 2.1. We list four entity types namely people, place, temporal expression and meetup theme type.

2.1.2 Components of the Knowledge Extraction Pipeline

As described previously, the pilot explores data that supports music historians' analysis of encounters between personalities in the musical world. Therefore, we rely on documentary sources such as biographies and select an open-access database such as Wikipedia as the primary data input. Figure 2.2 provides an overall picture of the knowledge extraction pipeline devised to process textual resources and build the knowledge graph of historical meetups. The pipeline consists of five main steps: *Data Collection, Cleaning & Segmentation, Entity Recognition, Coreference & meetup annotation, and Knowledge Graph Construction.*

In this deliverable, we address the software components developed for identifying and extracting the entities that are part of a historical meetup, such entities are grouped into people, place, time expression and meetup type. To obtain the list of targeted personalities, we queried the DBpedia SPARQL endpoint. We gathered a total of 33,309 biographies. Each biography was collected in text format from Wikipedia, using the identifier from DBpedia. In order

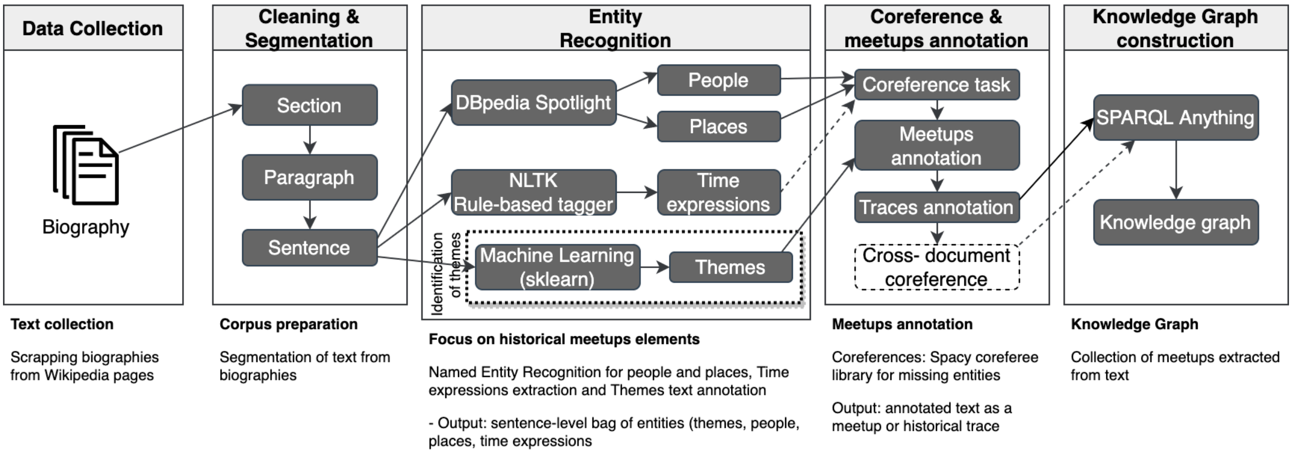

Figure 2.2: Historical MEETUPS - knowledge extraction pipeline.

to make experiments manageable, we randomly sampled a total of 1012 biographies that will be inspected in the following steps.

We start by collecting data that provides an overview of music personalities' lives, such as biographies in text format (`meetups-corpus-collection, meetups-data-cleaning`). In order to facilitate the location and classification of the entities, the text is organised in sentences, grouped by paragraphs and sections.

Next, we process the corpus to link entities such as places and people (`meetups-entity-recognition`), extracting time expressions (`meetups-time-extraction`) and classifying them according to meetup types (`meetups-themes`). The entity recognition and disambiguation process is performed at the sentence level. The output of this step is a bag of entities at the sentence level, grouped by paragraphs and sections.

Once the entities are extracted, we run the entity coreference resolution process (`meetups-coreference`) to find missing mentions of people and places in the document. This task is performed first at sentence level and then at paragraph level to minimise errors in the coreference task. This is an important consideration since mentions can be passed to many sentences in the same paragraph.

After the entity identification task is completed using coreference, we run the final process to find *historical meetups* within the text. This process is executed first at sentence level and uses the bag of entities identified by the previous components. If a sentence contains at least one entity for each entity type, then the component (`meetups-identification`) will annotate the text as a meetup. If there is a missing entity of any entity type, then the component will expand the search of the missing entity to adjacent sentences at paragraph level. The algorithm takes into consideration, that sentences refer to the same theme and the type of missing entity.

The last step is dedicated to building the MEETUPS Knowledge Graph[3] of historical meetups using as a framework the MEETUPS Ontology[4] [6, 7] and the data output from the previous step.

In this deliverable, we give details of the software tools for coreference (`meetups-coreference`) and meetup identification (`meetups-identification`). In addition, we expand on the previous Deliverable 4.3 [1] content to include software component updates since the last release. Figure 2.3 illustrates the set of software components developed to execute the different tasks of the knowledge extraction pipeline.

[3]MEETUPS Knowledge Graph repository: https://github.com/polifonia-project/meetups-knowledge-graph
[4]MEETUPS Ontology repository: https://github.com/polifonia-project/meetups-ontology

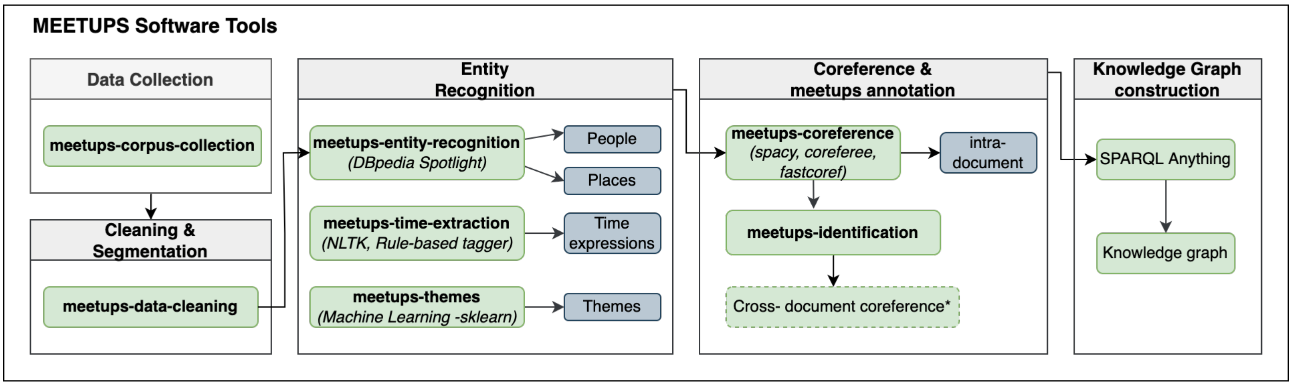

Figure 2.3: Historical MEETUPS - names of tools corresponding to the ID assigned in the Polifonia repository

2.2 Software for the identification of Historical Meetups

2.2.1 Entity coreference resolution

The outcome from the Entity Recognition step provides a set of entities identified at sentence level and then grouped by paragraph and section (see Figure 2.2). These results are the basis for identifying a historical meetup. Four entity types constitute a `historical meetup`: people, place, time expression and theme. When an entity type is missing there is the possibility that the text effectively does not provide information about a specific entity (therefore, the piece of text does not refer to a meetup) or that the entity was mentioned in the form of a noun phrase, or pronominal text, and therefore, not being recognised and linked to a DBpedia resource.

The Coreference Resolution (CR) task is an essential element for Natural Language Processing (NLP) applications. In this case, we refer to entity coreference resolution and define it as the process of finding all the mentions that point to the same entity in the text [8]. The main objective of CR is to find the mentions of entities as found in the text, for instance, in the form of noun phrases, named entities and pronouns.

The `meetups-coreference` component focuses on the resolution of mentions and identification of entities, in particular, identifying people and places mentions. This software supports the identification of missing entities during the entity recognition and linking task and leverages the opportunity to identify historical meetups. Furthermore, the software tool validates that these mentions refer to a named entity and links them to DBpedia or Wikipedia resources.

Importantly, this component boosts the identification of entities that are mentioned implicitly. The results then are used to identify historical meetups and as a consequence improve the data extracted and the number of historical meetups in our dataset.

It was developed using Python and uses the spaCy library *coreferee*[5]. One of the benefits of this library is that it can be integrated as a plugin in the spaCy's pipeline for Natural Language Processing. We use the spaCy's large English pipeline `'en_core_web_trf'`, this pipeline was trained using written web text and includes vocabulary, syntax and entities. Typically, execution of large pipelines is slower, however, it is compensated with improved accuracy (see spaCy documentation[6]).

The component takes as input text and runs the coreference resolution at paragraph level. The resolution may return mentions to dates, organisations, and names of works of art, among others. In our case, we concentrate on entities such as GPE, LOC and PERSON. Each entity (people or place) is then added to the bag of entities.

2.2.1.1 Summary

[5] Coreferee a spaCy library: https://spacy.io/universe/project/coreferee/
[6] spaCy models: https://spacy.io/models

ID	meetups-coreference
Type	Software
Title	MEETUPS - Entity coreference resolution
Credits	Alba Morales (OU) Enrico Daga (OU)
Description	This tool is part of the MEETUPS pilot and executes the coreference resolution task by identifying mentions of entities in the form of noun phrases, names, or pronominal text, particularly people and places. It leverages the identification of missing entities during the entity recognition and linking task and increases the possibility of identifying historical meetups. Furthermore, the tool validates that these mentions refer to the named entity and link them to DBpedia or Wikipedia resources. It was developed using Python and uses the spaCy library *coreferee*[7]. One of the benefits of this library is that it can be integrated as a plugin in the spaCy's pipeline for Natural Language Processing. The coreference task is executed at the paragraph level, as strategy a to minimise the number of errors and missing mentions. The coreferee library resolves coreferences within English text and uses a mixture of neural networks and programmed rules[8].
Source Code	https://github.com/polifonia-project/meetups_pilot
Documentation	README_meetups-coreference.md
Release	v0.2
Licence	Apache 2.0
Running Instance	Jupyter Notebook 05_Coreference.ipynb
Related Deliverables	Polifonia deliverable D4.3, Polifonia deliverable D4.4

2.2.1.2 Exploitability

This tool is part of the MEETUPS Knowledge extraction pipeline and processes text from music personalities' biographies in order to find entities mentioned in the form of noun phrases, named or pronominal text.

Other applications can use the Coreference Resolution (CR) component as it provides an algorithm that takes textual sources and produces as output a list of entity mentions, that includes an index of the first character of the mention and the link to the corresponding resource.

2.2.1.3 Future work

We consider it essential to test and evaluate the performance of other tools such as *fastcoref*[9] [9] and compare their performance since the meetups corpus contains biographies of artist from 1900s and most of these libraries are trained using data from news and in a small number of Wikipedia texts.

[7] Coreferee a spaCy library: https://spacy.io/universe/project/coreferee/
[8] Coreferee repository: https://github.com/richardpaulhudson/coreferee
[9] fastcoref repository: https://github.com/shon-otmazgin/fastcoref

2.2.2 Historical meetups identification

This tool is part of the MEETUPS pilot and processes text from music personalities' biographies to identify historical meetups. The software component gathers the output produced during the Entity Recognition and Coreference & meetups annotation steps (see Figure 2.3) and runs an automatic evaluation to identify if elements that define a historical meetup are present in a piece of text.

First, the tool examines each piece of text, at the sentence level, and evaluates if all the entity types are present. Sentences that comply with this requirement are automatically annotated as a historical meetup. All the sentences that have one or more entity types missing are annotated temporarily as a historical trace. The next step is to search for complementary information within the same paragraph and potentially identify more historical meetups. The algorithm searches for missing entities in previous sentences within the same paragraph, the aim is to group two or more consecutive sentences that refer to the same event and among them feed all the required entity types of a historical meetup. These sentences should refer to the same type of encounter in order to be considered and not have been previously annotated as a historical meetup.

The historical meetup identification tool was developed using Python and Jupyter Notebook. As input, it uses the bag of entities obtained from the Entity Recognition and Coreference steps. The output is a corpus that contains the text (typically a sentence or a set of sentences), and the list of entities that account for a meetup. Table 2.2 displays additional information about the information extracted. We annotated a total of 3186 historical meetups, found in 743 different biographies of personalities in the musical world. The results are stored in CSV files, grouped by biographies. The corpus is used later to build the MEETUPS KG.

	Total
Biographies	743
Meetups	3186
People mentions	8569
Places mentions	1581
Time expressions	4777

Table 2.2

2.2.2.1 Summary

ID	meetups-identification
Type	Software
Title	MEETUPS - Historical meetups identification
Credits	Enrico Daga (OU)
Description	MEETUPS Historical meetup identification tool was developed using Python and Jupyter Notebook. As input, it uses the bag of entities obtained from the Entity Recognition and Coreference steps. The output is a corpus that contains the text (typically a sentence or a set of sentences), and the list of entities that account for a meetup. The results are stored in CSV files, grouped by biographies. The corpus is used later to build the MEETUPS KG. The software component gathers the output produced by the tools of Entity Recognition and Coreference & meetups annotation steps (see Figure 2.3) and runs an automatic evaluation to identify if elements that define a historical meetup are present in a piece of text.
Source Code	https://github.com/polifonia-project/meetups_pilot

Documentation	README_hm-identification.md
Release	v0.2
Licence	Apache 2.0
Running Instance	Jupyter Notebook 06_MeetupsIdentification.ipynb
Related Deliverables	Polifonia deliverable D4.3, Polifonia deliverable D4.4

2.2.2.2 Exploitability

This component is a key piece for the identification of historical meetups. It allows the automatic annotation of text as an encounter. It also introduces the concept of `historical trace` as the annotation used to identify text that contains just enough information to answer partially some of the MEETUPS Competency Questions. For instance, a piece of text that indicates a musical encounter between two artists in a particular city but without reference to the time the event took place.

This tool makes it practical to identify the text that refers to a historical meetup and lists the constituent elements (entity types) of a meetup. Crucially, the output produced by the execution of the component is used to build the MEETUPS Knowledge Graph and visualised using the MEETUPS Web tool[10].

2.2.2.3 Future work

We plan to expand the rules implemented for the identification of historical meetups. In fact, the main limitation of this module is that sentences that do not have all the entity types present are immediately annotated as historical traces; these information classified as traces should be validated to identify how much information can be provided to answer the Competency Questions formulated as part of the MEETUPs Pilot design and requirements process.

Future work is required to assess the quality of the results obtained, these assessment demands experts' knowledge in the musical history domain.

Furthermore, future work will also include a module to identify cross-document coreference. The aim is to link historical meetups that describe the same or similar events in two or more biographies. By building connections between events, we expect to enrich the artists' biographies and, provide enough information to scholars in analysing social interactions at scale.

2.2.3 Update: Identification of people and places

This component was released as part of the D4.3 Deliverable [1] and has been updated to improve the quality of the captured entities. Particularly, this update includes improvements to implement some of the findings of the survey carried out to evaluate the quality of the results and published in [7] and we summarise in what follows:

- The cases in which people were incorrectly identified mainly pertain to homonyms names. For example, people's names are used to name prizes, and festivals, among other cases. We introduced further validation to include querying DBpedia or Wikidata to determine if the entity is a person (for example, has a date of birth) and the entity type (for example, if it refers to a person or musician).
- We include validation of people entities to minimise the number of named entities that are incorrectly identified. For instance, named entities referring to people were wrongly linked to contemporary artists (born in the last 50 years) instead of linking them to artists who lived in Europe between c. 1800 and c. 1945 (the aim of our research). We validate the quality of the entity linked by comparing the date of birth of the linked resource to the actual period of the artist's biography. Furthermore, entities are listed even when it does not have a resource linked to them.

[10] MEETUPS web tool repository: https://github.com/polifonia-project/meetups-application

A survey was carried out to evaluate the accuracy of the results. We sampled 100 sentences and asked participants (three in total, all English speakers) to read and manually annotate people and place mentions. They annotated a total of 94 sentences, identifying 78 people entities and 36 place entities. We compared these manual annotations with the results of the automatic extraction. Our tool accurately extracts and links 83% of people entities and 85% of place entities; in both cases, these results show an improvement in the entity recognition process compared to the previous version of the component.

2.2.3.1 Summary

ID	meetups-entity-recognition
Type	Software
Title	MEETUPS identification of people and places
Credits	Alba Morales (OU) Enrico Daga (OU)
Description	This tool uses DBpedia Spotlight to identify and annotate possible entity mentions from input text. This is an essential process to identify two of the three main elements that define a meetup: people (who participated) and place (where). It uses as input the corpus of music personalities generated by the MEETUPS cleaning component https://github.com/polifonia-project/meetups_pilot/blob/main/01_CleaningText.ipynb. The implementation is divided into two Jupyter notebooks: The first notebook is in charge of querying DBpedia Spotlight, processing the responses (JSON format) and storing responses locally. The second notebook uses the responses from DBpedia Spotlight to capture data about people (identified as DBpedia:Person) and places (identified as DBpedia:Place).
Source Code	https://github.com/polifonia-project/meetups_pilot
Documentation	README_people_places_identification.md
Release	v0.3
Licence	Apache 2.0
Running Instance	Jupyter Notebook 02_queryDbpedia.ipynb 02_Identify_PP.ipynb
Related Deliverables	Polifonia deliverable D4.3

2.2.3.2 Exploitability

This tool is part of the KG construction pipeline. The entity recognition task is in charge of identifying and extracting two out of four components of a historical meetup: participants or people involved (who) and places (where). The data generated by this component will be used to build a knowledge graph of historical meetups.

2.2.3.3 Future work

We consider as future work the inclusion of other databases such as Wikidata or MusicBrainz for the entity linking task.

2.2.4 Update: Identification of temporal knowledge

In this section, we describe the update of the component to identify temporal expressions from text. As described in detail in the previous D4.3 Deliverable [1], this software component is based on research by [10] and SynTime software[11]. According to the authors, SynTime implements a three-layer system that recognises time expressions using syntactic token types and general heuristic rules. Unlike SynTime, a JAVA implementation, our software was developed using Python and the NLTK Toolkit. The tool classifies temporal expressions into one of the three types of expressions: time range (e.g., from 1938 to 1987, to 1900, until 1987), time point (e.g., exact date, 23/03/1294), and time reference (e.g., incomplete dates such as 19 April, two weeks, later this year, in 1990).

The updated component described in this section includes improvements regarding the identification of temporal expressions. Particularly, we address the findings concerning the number of missing and incorrectly identified temporal expressions discussed in [7] and [11].

In the case of missing expressions, this release extends the heuristic and reasoning rules in order to enhance syntactic token type tagging. Regarding incorrectly identified time expressions, we review the heuristic rules to include Python-specific regex expressions (SynTime is a Java development).

A survey was carried out to evaluate the accuracy of the results. We sampled 100 sentences and asked participants (three in total, all English speakers) to read and manually annotate temporal expressions. In the end, participants annotated 94 sentences and identified 36 temporal expressions. We compared these manual annotations with the results of the automatic extraction. Our tool accurately extracts 88% of temporal expressions; in comparison to results obtained using the previous release of the tool [11], these results show a substantial improvement. Regarding missing temporal expressions, we effectively minimised the number of missing mentions to only 5.5% of the total.

An essential aspect of this release is the normalisation of the temporal expressions, the objective is to recognise those expressions from unstructured text and parse them to a standard format. We follow the International Standard for representation of dates and times, specifically, ISO 8601 and implement the *YYYY-MM-DD* format. We investigated available tools and selected two Python libraries that support the tasks at hand.

- `dateutil`[12] is a powerful Python library that supports generic parsing of dates (ISO 8601 by default) in almost any string format.
- `approx-dates`[13] is a Python library that creates a full date or a partial date from a formatted ISO 8601 string and implements a series of functions to obtain approximate dates. For instance, given a year (1945), it automatically is formatted to the earliest date (1945-01-01) and the latest date (1945-12-31).

The `dateutil` library proved to be highly efficient in parsing temporal expressions identified as *time points* with 100% of them being normalised into a standard format. For instance, the date *'2 June 1857'* is normalised as '1857-06-02'. To normalise temporal expressions of the *time range* type, we use the `dateutil` and `approx-dates` libraries in combination. On average, 50 to 70 % of the time range expressions found in each biography are normalised. For instance, expressions like *'1914 and 1925'* are parsed as `start`: '1914-01-01' and `end` '1925-12-31'. For *time reference* expressions, we use POS tags and a series of heuristic rules combined with the libraries `dateutil` and `approx-dates`. We normalise an average of 50-60% of temporal expressions found in each biography. For instance, an expression such as *'the 1960s'* is normalised as `start`: '1960-01-01' and `end` '1969-12-31'.

2.2.4.1 Summary

ID	meetups-time-extraction
Type	Software
Title	MEETUPS - Identification of temporal knowledge

[11] SynTime repository: https://github.com/zhongxiaoshi/syntime
[12] dateutil library: https://dateutil.readthedocs.io/en/stable/index.html
[13] approx-dates library: https://pypi.org/project/approx-dates/

Credits	Alba Morales (OU) Enrico Daga (OU)
Description	This tool is part of the MEETUPS pilot and processes text from music personalities' biographies to find time expressions. This is part of the process to identify one out of the four elements that define a meetup: the date or moment in time when it happened (when), along with data of the people involved (who), the place (where) and the event that took place (what) complete the historical meetup information. It uses the Python NLTK library and a set of heuristic rules to identify and annotate temporal knowledge from text. This implementation is a rule-based Time Expression recognition tagger based on research by [10] and SynTime software (https://github.com/zhongxiaoshi/syntime). This component classifies time expressions into three types of expressions: time range (e.g., from 1873 to 1875, from 1987, until 1845), time point (e.g., exact date, 23/03/1294), and time reference (e.g., incomplete dates such as 19 April, two weeks, later this year). Importantly, our tool provides a feature to normalise temporal expressions into the ISO 8601 standard and implement the *YYYY-MM-DD* format. We use a combination of heuristic rules and functions provided by the Python libraries `dateutil` and `approx-dates`.
Source Code	https://github.com/polifonia-project/meetups_pilot
Documentation	README_time_expressions.md
Release	v0.2
Licence	Apache 2.0
Running Instance	Jupyter Notebook 03_Identify_TimeE.ipynb
Related Deliverables	Polifonia deliverable D4.3

2.2.4.2 Exploitability

This tool is part of the MEETUPS pilot and processes text from music personalities' biographies to find temporal expressions.

Our tool classifies temporal expressions into one of the three types of expressions: time range (e.g., from 1873 to 1875, from 1987, until 1845), time point (e.g., exact date, 23/03/1294), and time reference (e.g., incomplete dates such as 19 April, two weeks, later this year).

Another important aspect of this update is the normalisation feature that processes as input temporal expressions and parses them into standard date format. Different types of temporal expressions are now transformed into data that can be queried. The data generated is used for building the MEETUPS Knowledge Graph.

2.2.4.3 Future work

Future work includes evaluating the accuracy of the extracted time expressions by comparing the tool performance against state-of-the-art methods such as SynTime on well-known datasets such as TimeBank and WikiWars.

Regarding temporal expression normalisation, further work will need to be done to expand the normalisation of `time reference` expressions type, since it proved to be challenging to parse these expressions into approximate dates.

2.2.5 Update: Identification of themes

In this section, we describe the component in charge of analysing text and extracting the type of encounter expressed according to a set list of classes. The meetup type describes the *reason/purpose* for an encounter. According to scholars and researchers of the Music Department at The Open University, meetups can be grouped as Business meetings and career (e.g., contract, retirement), Personal life (e.g., born, divorce, family), Coincidence (e.g., meet, find, discover), Education (e.g., learn, teach, conservatoire), Public celebration or Music-making (e.g., play, produce) [7].

This component was released as part of the D4.3 Deliverable [1] and has been updated to include improvements regarding the identification of meetup type. Particularly, we address the findings concerning the number of incorrectly identified themes after a survey was carried out to evaluate the accuracy of the results.

The previous release of the MEETUPS identification of themes component was developed using Python and Jupyter Notebook. It uses Python SKLEARN and a set of Machine Learning algorithms to classify sentences according to the established type of events.

This new component uses the latest developments in the field of artificial intelligence and generative AI systems. Particularly, we use the Large Language Model tool ChatGPT to classify pieces of text according to the main topic/theme they describe. We followed a zero-shot learning approach, where the tool takes a piece of text, and the list of classes and has to evaluate which of the classes better describes the content of the sentence.

This component was developed using Python and the OpenAI API to interrogate the GPT3 with the following prompt:

```
1  You are a knowledge classification system that annotates sentences according to their main topic.
2  Respond in JSON format with one of the following keys:
3  thm_type_1, thm_type_2, thm_explanation_1 and thm_explanation_2.
4  The value for thm_type_1 should be the first most probable topic from one of the following keys:
5  ['Music making', 'Business meeting', 'Personal life', 'Coincidence', 'Public celebration',
6  'Education']
7  The value for thm_type_2 should be the second most probable topic from one of the following keys:
8  ['Music making', 'Business meeting', 'Personal life', 'Coincidence', 'Public celebration',
9  'Education']
10 The value for thm_explanation_1 should be a very short explanation for the topic in thm_type_1.
11 The value for thm_explanation_2 should be a very short explanation for the topic in thm_type_2.
```

The prompt is designed to:

1. Set the context of the task: *a classification system* (see line 1).
2. Set the expected output format: JSON format (lines 2 and 3).
3. Explain the expected output: instructing the tool on the task of classifying the text according to the set list of classes(lines 4 to 11).

For each sentence, we gather the two most probable topics of the text. For instance, the text: *'His father, William Henry Elgar (1821–1906), was raised in Dover and had been apprenticed to a London music publisher.'* will be classified as follows:

```
{'thm_type_1': 'Music making',
 'thm_explanation_1':
 'The sentence mentions a music publisher, indicating a connection to music making.',
 'thm_type_2': 'Personal life',
 'thm_explanation_2':
 'The sentence also provides information about the personal life of Elgar's father.'}
```

We carried out a survey to evaluate the accuracy of the results. We sampled 100 sentences and asked participants (three in total, all English speakers) to read and manually annotate the theme of the encounter using the set of classes listed previously. In the end, participants annotated 83 sentences and identified 83 theme types.

We compared these manual annotations with the results of the automatic extraction using the Machine Learning (ML) approach, the previous component, and the Large Language Model (LLM), the new component. Table 2.6 displays the results in terms of Precision. Clearly, using LLM tools leverages the classification of text, increasing its precision to 85% (on average).

		Machine Learning (ML) prediction		LLM prediction	
	# Sentences	**Precision @ 1**	**Precision @ 2**	**Precision @ 1**	**Precision @ 2**
Edward Elgar	56	0.34	0.64	0.45	0.73
Yehudi Menuhin	11	0.36	0.45	0.55	0.82
Clara Butt	16	0.69	0.75	0.88	1
	83	0.46	0.62	*0.62*	*0.85*

Table 2.6: Classification results. Machine Learning approach and LLM Precision results @ 1 and 2

2.2.5.1 Summary

ID	meetups-themes
Type	Software
Title	MEETUPS - Identification of themes
Credits	Alba Morales (OU) Enrico Daga (OU)
Description	This tool is part of the MEETUPS pilot and processes text from music personalities' biographies to find encounter types. It previous release uses "sklearn" and a set of Machine Learning algorithms to classify sentences according to the established type of events. In this new release, we make use of the latest developments in the AI field and experiment with Large Language Models, particularly, we use this technology to classify sentences automatically according to their topic and assign the corresponding class. The tool extracts information from one of the four elements defining a meetup: the type of encounter (what). Encounter type, along with data of the people involved (who), the place (where) and the time it took place (what), complete the historical meetup information.
Source Code	https://github.com/polifonia-project/meetups_pilot
Documentation	README_identification_themes.md
Release	v0.2
Licence	Apache 2.0
Running Instance	Jupyter Notebook 03_Identify_TimeE.ipynb
Related Deliverables	Polifonia deliverable D4.3

2.2.5.2 Exploitability

In this new release, we make use of the latest developments in the AI field and experiment with Large Language Models, particularly, we use this technology to classify sentences automatically according to their topic and assign the corresponding class.

We designed a prompt that can be used to classify other datasets according to classes of interest. For instance.

2.2.5.3 Future work

Future work includes evaluating the accuracy of the extracted themes with experts and expanding the sample of texts used for the experiments. Another potentially fruitful avenue for future research includes the analysis of prompt engineering and how designing different prompts could impact the classification task. For instance, providing a more detailed description of the context (for instance, situated in the music domain) or even avoiding providing a context.

3 Polifonia Music Lexicon Annotator

The Polifonia Music Lexicon Annotator [12] is a software component that allows the identification of musical concepts in a text. This module takes a text as input and annotates the content words in it with word sense information. Word senses are taken from WordNet [13], employing the Polifonia Lexicon, developed within the project and released with Deliverable 4.1 [14], to identify if a particular word sense is related to music or not. The disambiguation algorithm used to associate word senses with word form is based on the most frequent sense heuristic, computed on the SemCor dataset [15]. The output annotation is in JSON and includes tokenization, lemmatization and part of speech tagging, word senses and a flag that indicates if a particular concept in a sentence is related to music. This module has not undergone major changes since its first release, documented in Deliverable 4.3.

3.1 Summary

ID	polifonia-musical-lexicon-annotator
Type	Software
Title	Polifonia Musical Lexicon Annotator
Credits	Rocco Tripodi (UNIBO) Arianna Graciotti (UNIBO)
Description	This software provides an easy-to-use tool for recovering musical concepts in a text, taking the Polifonia Lexicon, based on WordNet as a reference
Source Code	https://github.com/polifonia-project/MusicalLexiconAnnotator
Documentation	README.md
Release	v0.0.1
Licence	CC-BY
Bibliography	Rocco Tripodi et. al, "D4.1: Plurilingual corpora containing source texts in English, French, Spanish and German (v1.0)" (2021).
Related Deliverables	Polifonia deliverable D4.3

3.2 Exploitability

This module lends itself very well to fast annotation of textual documents. It allows the identification of a large corpus of only documents that contain key concepts related to music. This feature makes it very useful for the study of musical terminology or for the selection of documents relevant to certain aspects of music.

This module has been employed to annotate the Polifonia Textual Corpus with Musical Heritage-specific annotations based on the Polifonia Lexicon. The resulting annotations are accessible to linguists, musicologists, scholars, music professionals and others through a custom GUI[1], implemented by leveraging the Polifonia Corpus Annotation APIs. The GUI's functionalities offer the possibility of querying the corpus through advanced semantic, lexicon and music-specific information encoded in the annotations.

[1] https://polifonia.disi.unibo.it/corpus/

3.3 Future Work

We plan to integrate into the repository different disambiguation algorithms in order to have better performance, especially in the music domain. In fact, the main limitation of this module is that if the disambiguation is wrong the identification of the resulting music concept will be wrong. We plan also to extend the language coverage, including all the languages covered by the Polifonia Textual Corpus, i.e., Dutch, English, German, French, Italian and Spanish.

4 Search expansion using Large Language Models (LLM)

This software component was developed with the aim of supporting the identification of implicit themes in text and takes as reference the documentary evidence benchmark [16, 17, 18].

This component uses as input one general theme (an entity, for instance, childhood) and uses Large Language Model (LLM) tools such as ChatGPT[1] to generate a list of closely related words. These words are then used to build an expanded query that will retrieve listening experiences about the specified theme. The objective of this component is to accelerate the retrieval of listening experiences related to a theme of interest to researchers.

Current search tools use as input one or more keywords, typically, these keywords have to be explicitly mentioned in the document (for instance, in the description of the listening experience) in order to be retrieved by the tool. Furthermore, the user may have to interact several times using related words. With our software component, we intend to expand the search automatically by including not only related words but terms that define an entity of interest and represent implicit themes, therefore, effectively retrieving documents that are closely related to a given theme.

The work towards expanding search tools for supporting the Listening Experience Database (LED) annotation was motivated by previous work presented in the D4.3 Deliverable [1] related to the documentary evidence benchmark and its curated list of listening experiences related to childhood (of relevance to the CHILD pilot).

4.1 Summary

ID	child-search-expansion
Type	Software
Title	CHILD - Search expansion using LLMs
Credits	Jason Carvalho (OU) Alba Morales (OU) Enrico Daga (OU)
Description	This software component was developed with the aim of supporting the identification of implicit themes in text and takes as reference the documentary evidence benchmark. This component uses as input one general theme (an entity, for instance, childhood) and uses Large Language Model (LLM) tools such as ChatGPT to generate a list of closely related words. These words are then used to build an expanded query that will retrieve listening experiences about the specified theme.
Source Code	https://github.com/polifonia-project/child-search-expansions
Documentation	README.md
Release	v0.2.1
Licence	Apache 2.0
Bibliography	Jason Carvalho, Alba Morales Tirado, Enrico Daga (2023). polifonia-project/child-search-expansions: v0.2.1 (v0.2.1). Zenodo. https://doi.org/10.5281/zenodo.8322490

[1] ChatGPT description https://en.wikipedia.org/wiki/ChatGPT

Related Deliverables	Polifonia deliverable D4.4

4.2 Exploitability

The initial software component that has been developed works around a basic keyword expansion concept, using an LLM to take a single keyword and generate a number of synonyms. In this example, we are accessing ChatGPT from a Python script via their API interface, issuing it with the following prompt:

"Expand each of the terms listed below into synonyms and related terms. Only use single words for the related terms or synonyms. Display the new terms in a JSON structure with the original terms as a key and the list of terms as an array attribute".

An example response from ChatGPT to the prompt above, when passed the keyword *childhood* would be:


```
{
  "childhood": [
    "adolescence",
    "youth",
    "infancy",
    "kidhood",
    "early years",
    "innocence",
    "juvenility",
    "minority",
    "toddlerhood",
    "prepubescence",
    "puppyhood",
    "nursery"
  ]
}
```


The above set of keywords is shown as an example, but it should be noted that if the ChatGPT prompt shown above is submitted a number of times, the results differ slightly on each execution, both in terms of the actual words returned and the number of words returned. This presents both a challenge and an opportunity for effective search expansion and is behaviour that can be adjusted via prompt tuning and API settings adjustments.

For example, the prompt could be amended to ask the LLM to always return exactly 15 results. These variations have a significant effect on the generated SPARQL query and resulting triples returned from the knowledge graph. In running this experiment six times, we saw the number of knowledge graph entries returned vary from 21 to 168.

Initially, we experimented with also asking the LLM to build dynamic SPARQL queries. ChatGPT showed some early and promising success in generating working queries when given an example query on our knowledge graph that it could extrapolate and infer property names from. However, the variability and unpredictability of results gave occasional invalid queries, therefore, the decision was made to simply build the queries ourselves in a script, from the list of keywords.

These preliminary results are the starting point towards building a tool that accelerates the retrieval of documents according to a specific theme. Further experiments include an analysis of the steps taken to generate the list of words; reviewing the chain-of-though process by the LLM. Another alternative could be to provide context about the task, by including examples of results expected, in this case using the documentary evidence benchmark.

4.3 Future work

Future work includes experiments in order to fine-tune the LLM prompts employed to generate the list of expanded words. Prompt engineering fine-tuning has proved to be useful in obtaining more accurate results, however, the specification of prompts may affect the number of expansions generated. It is important to experiment with a larger number of settings.

Further work is needed to escalate the search and include, for instance, the automatic search of entities in related triples. For example, searching for listening experiences that *"took place in Italy"* will generate an expanded query that not only searches for documents that contain 'Italy' in the text but automatically identifies the type of entity we are looking for (e.g., `led:Location`), and returns the type of triple that should be part of the query (e.g., `?s led:hasLocation ?value`, where `?value` is 'Italy').

5 FAIR Protocol

The following software component have been released or updated as a result of the work described in this deliverable:

1. MEETUPS
   a) Intra-document coreference resolution component
   b) Identification of Historical Meetups component
   c) Software for entity recognition, it includes people, places, temporal expressions and themes resolution.
2. MUSICBO
   a) Polifonia lexicon annotator
3. CHILD
   a) Search expansion using Large Language Models (LLMs

Our work in this chapter complies with the FAIR[1] principles and is in accordance with the Polifonia First Data Management Plan (D7.1).

We have implemented a number of actions to ensure that these outputs are in compliance with the FAIR Guiding Principles for scientific data management and stewardship. Being integrated as part of the Polifonia Ecosystem, all software, data and metadata produced are stored and made available in order to ensure its use beyond the project deadline. For instance, all the resources are stored on GitHub, which guarantees the persistence of such resources. Furthermore, all GitHub repositories are published on Zenodo and have an associated DOI, useful for citing and localisation.

The results of the knowledge extraction pipeline for historical meetups and the results obtained from this validation are part of published work [7] and new features will be part of future published work.

[1] https://www.go-fair.org/fair-principles/

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